

CS605PC: ARTIFICIAL INTELLIGENCE LAB

B.Tech. III Year II Sem.

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Course Objectives:

- Become familiar with basic principles of AI toward problem solving, knowledge representation, and learning.

Course Outcomes:

- Apply basic principles of AI in solutions that require problem solving, knowledge representation, and learning.
- Implement simple PEAS descriptions for given AI tasks.
- Develop programs to implement simulated annealing and genetic algorithms.
- Demonstrate the ability to solve problems using searching and backtracking.
- Ability to implement simple reasoning systems using either backward or forward inference.
- apply appropriate algorithms for solving given AI problems.

LIST OF EXPERIMENTS:

Write Programs to implement the following using Python.

1. Breadth First Search
2. Depth First Search
3. Tic-Tac-Toe game
4. 8-Puzzle problem
5. Water-Jug problem
6. Travelling Salesman Problem
7. Tower of Hanoi
8. Monkey Banana Problem
9. Alpha-Beta Pruning
10. 8-Queens Problem

TEXT BOOK:

1. Artificial Intelligence a Modern Approach, Third Edition, Stuart Russell and Peter Norvig, Pearson Education.

REFERENCE BOOKS:

1. Artificial Intelligence, 3rd Edn, E. Rich and K. Knight (TMH)
2. Artificial Intelligence, 3rd Edn., Patrick Henny Winston, Pearson Education.
3. Artificial Intelligence, Shivani Goel, Pearson Education.